Session 16: Advanced Data Retrieval

Presented by

Ravindu Sooriyaarachchi

Todays Goal

- VLOOKUP
- XLOOKUP
- Pivot Tables
- Slicers
- Goal Seek

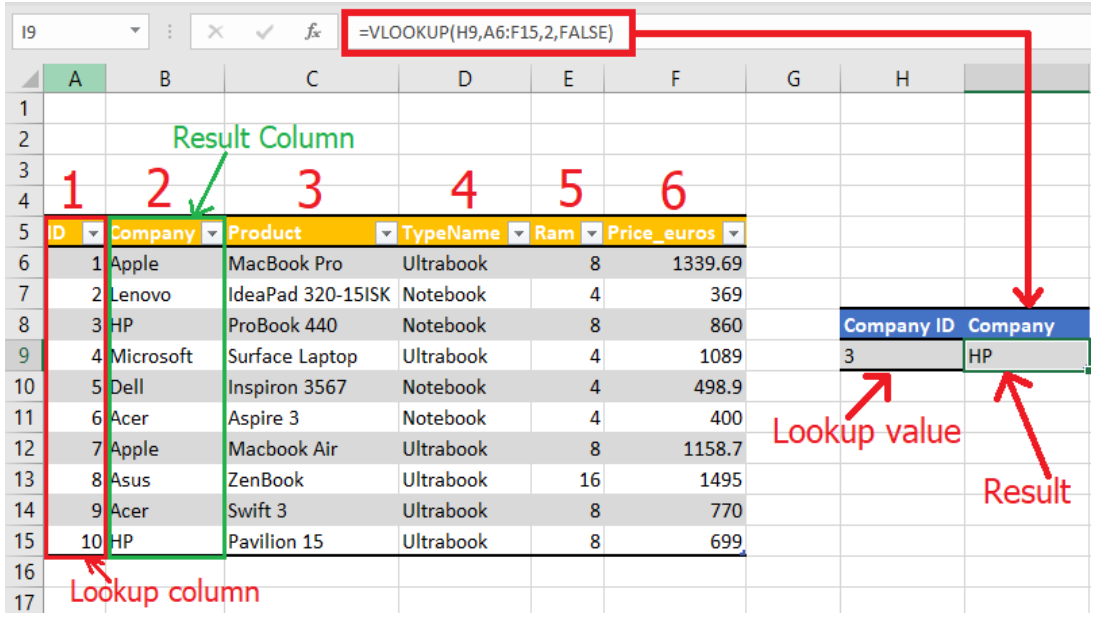
Finding the Needle in the Haystack

- Imagine a database of 10,000 employees.
- You need the salary of ID #4502.
- Scrolling takes too long.
- Solution: Excel Lookup Functions.
- It acts like a digital phonebook.

How VLOOKUP Works

- Requires 4 pieces of information:

1. What are you looking for? (e.g., ID 4502)
2. Where is the table? (e.g., A1:D1000)
3. Which column has the answer? (e.g., Column 4)
4. Exact Match: Always type FALSE.

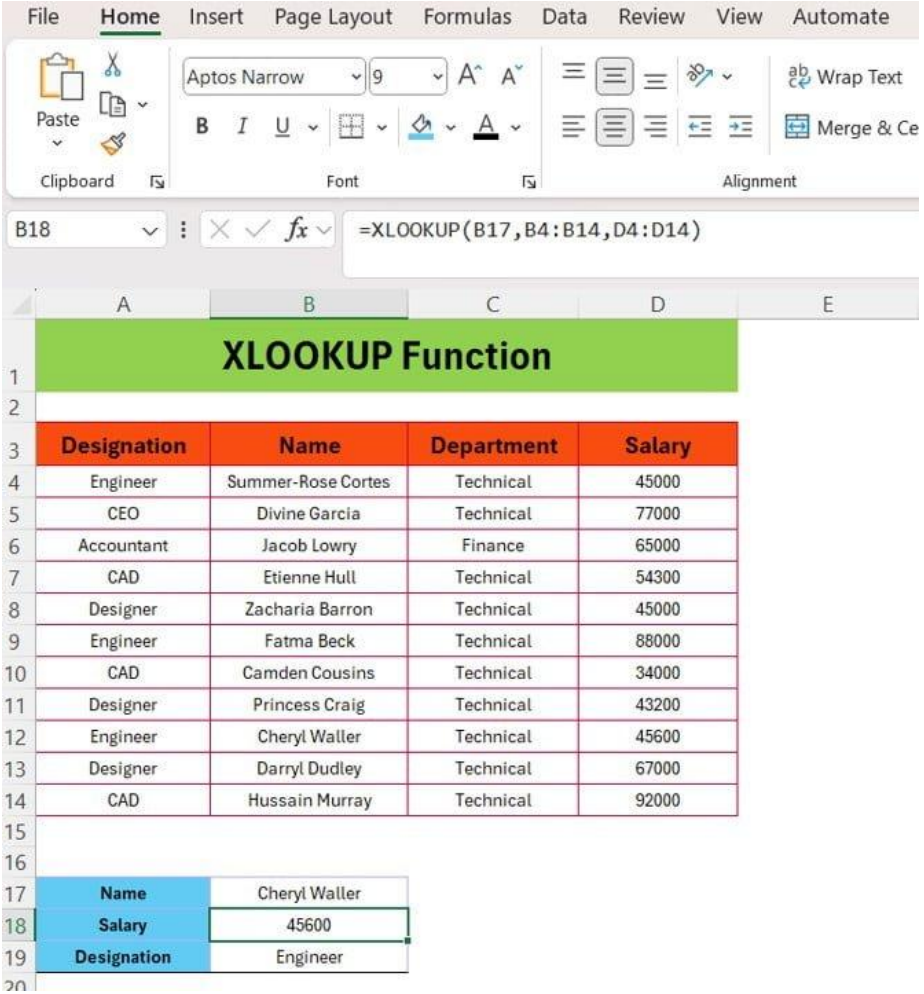


[VLOOKUP Worksheet]

** Practice Dataset provided by Kevin Stratvert (YouTube).*

Pro Tip: XLOOKUP

- VLOOKUP is famous, but old.
- XLOOKUP is the modern replacement (Office 365).
- Much easier to write.
- Cannot break when you insert new columns.
- Formula:
- =XLOOKUP(lookup_value, lookup_array, return_array)

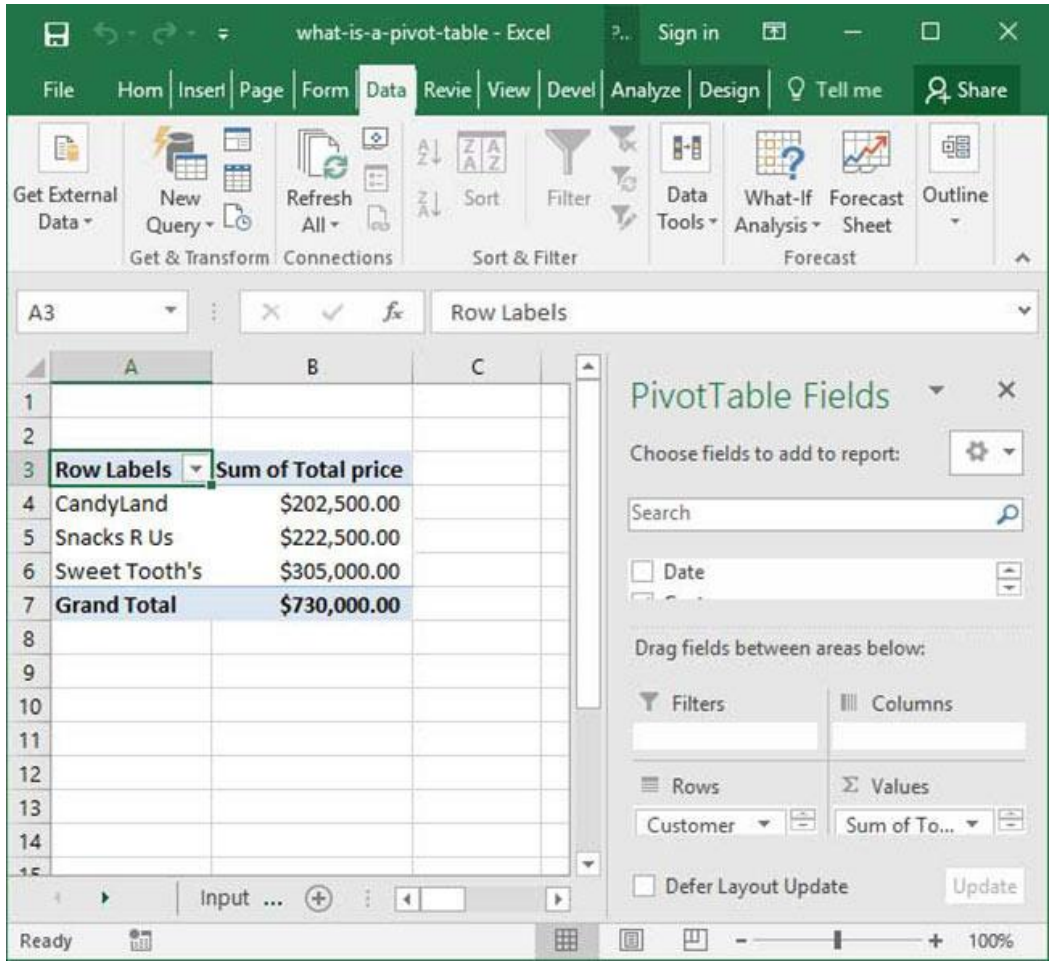


Designation	Name	Department	Salary
Engineer	Summer-Rose Cortes	Technical	45000
CEO	Divine Garcia	Technical	77000
Accountant	Jacob Lowry	Finance	65000
CAD	Etienne Hull	Technical	54300
Designer	Zacharia Barron	Technical	45000
Engineer	Fatma Beck	Technical	88000
CAD	Camden Cousins	Technical	34000
Designer	Princess Craig	Technical	43200
Engineer	Cheryl Waller	Technical	45600
Designer	Darryl Dudley	Technical	67000
CAD	Hussain Murray	Technical	92000

Name	Cheryl Waller
Salary	45600
Designation	Engineer

[XLOOKUP Worksheet]

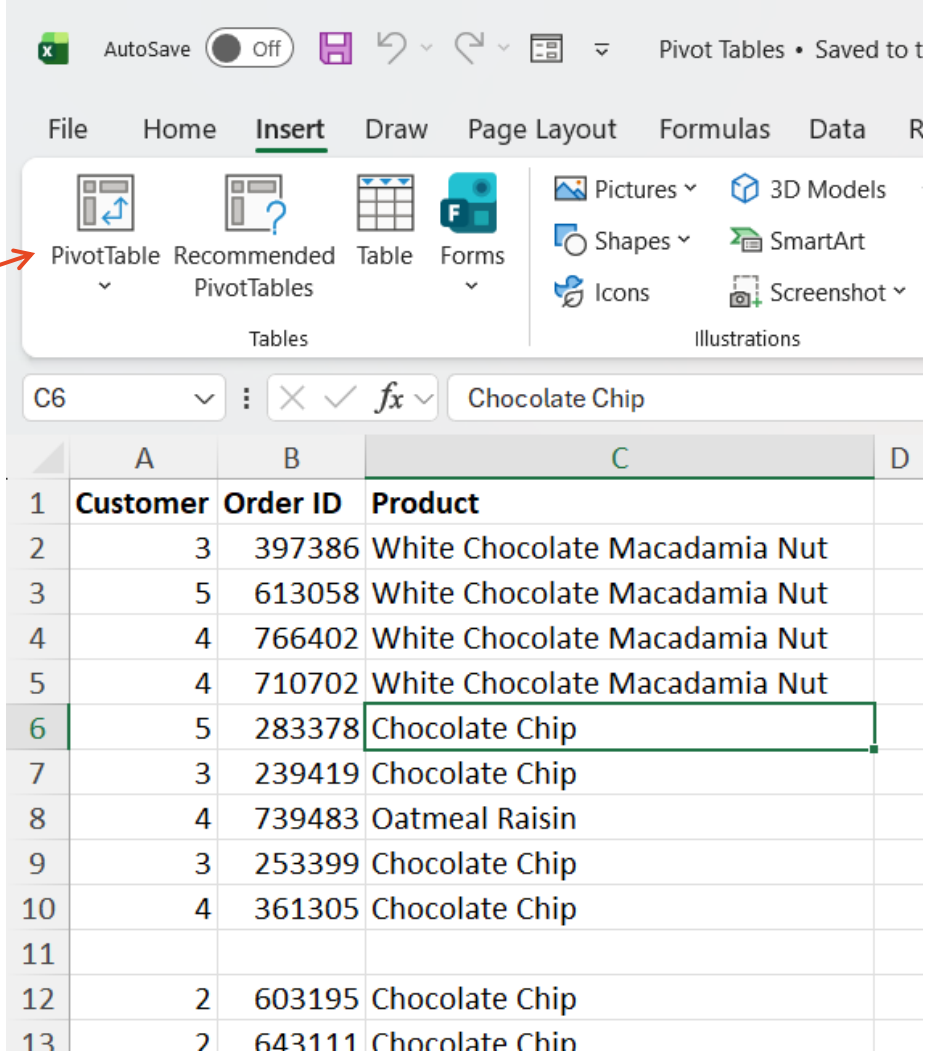
** Practice Dataset provided by Kevin Stratvert (YouTube).*



Pivot Tables

- Standard math (SUM, COUNT) is slow.
- Pivot Tables summarize 100,000 rows instantly.
- Zero formulas required.
- Just drag and drop!
- Goal: Turn raw data into business reports.

Building Your First Pivot



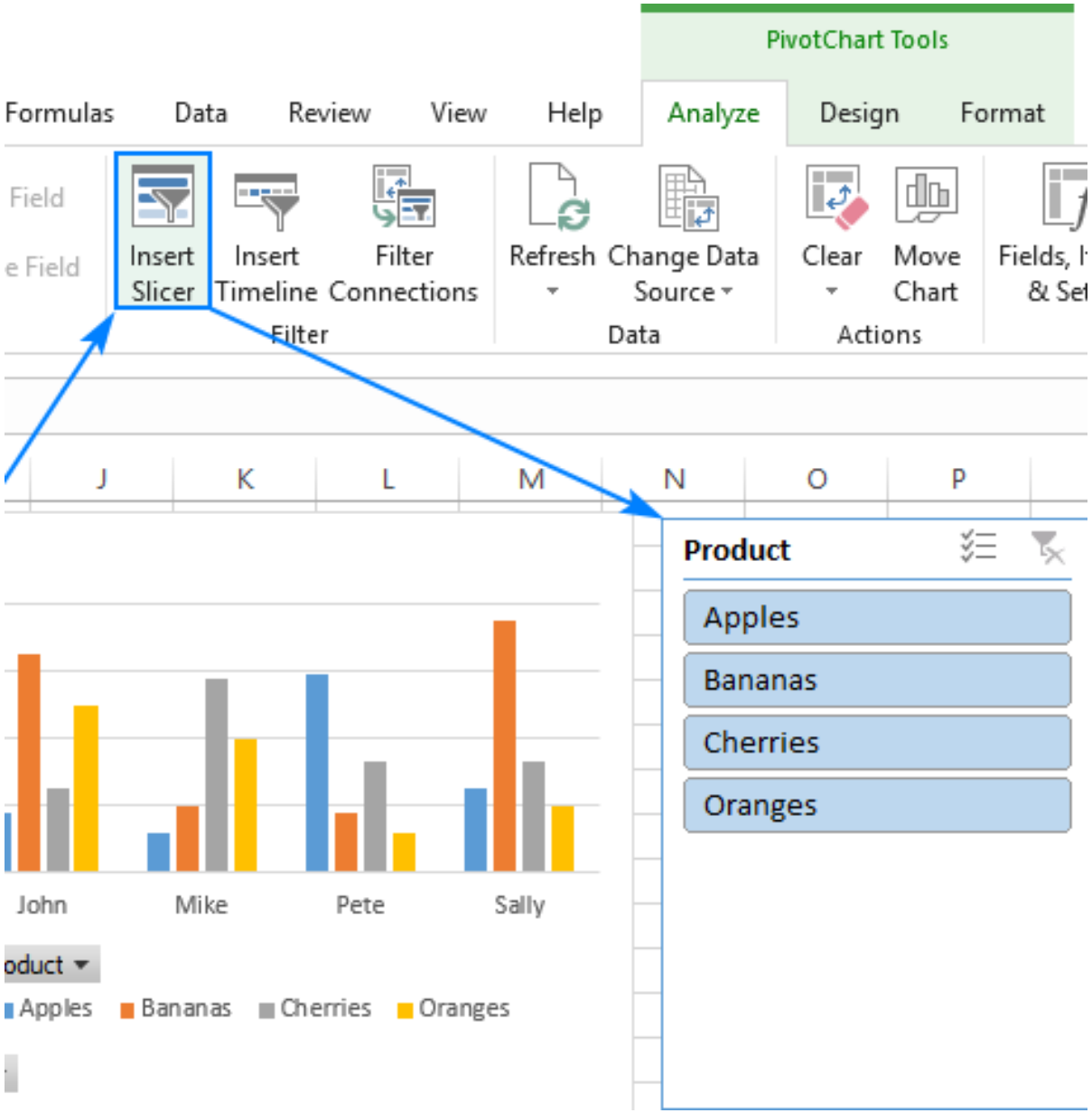
- Step 1: Click anywhere inside your data.
- Step 2: Insert -> PivotTable.
- Step 3: Choose "New Worksheet".
- Step 4: Drag fields (e.g., 'Department' to Rows, 'Sales' to Values).

Pivot Table Worksheet

** Practice Dataset provided by Kevin Stratvert (YouTube).*

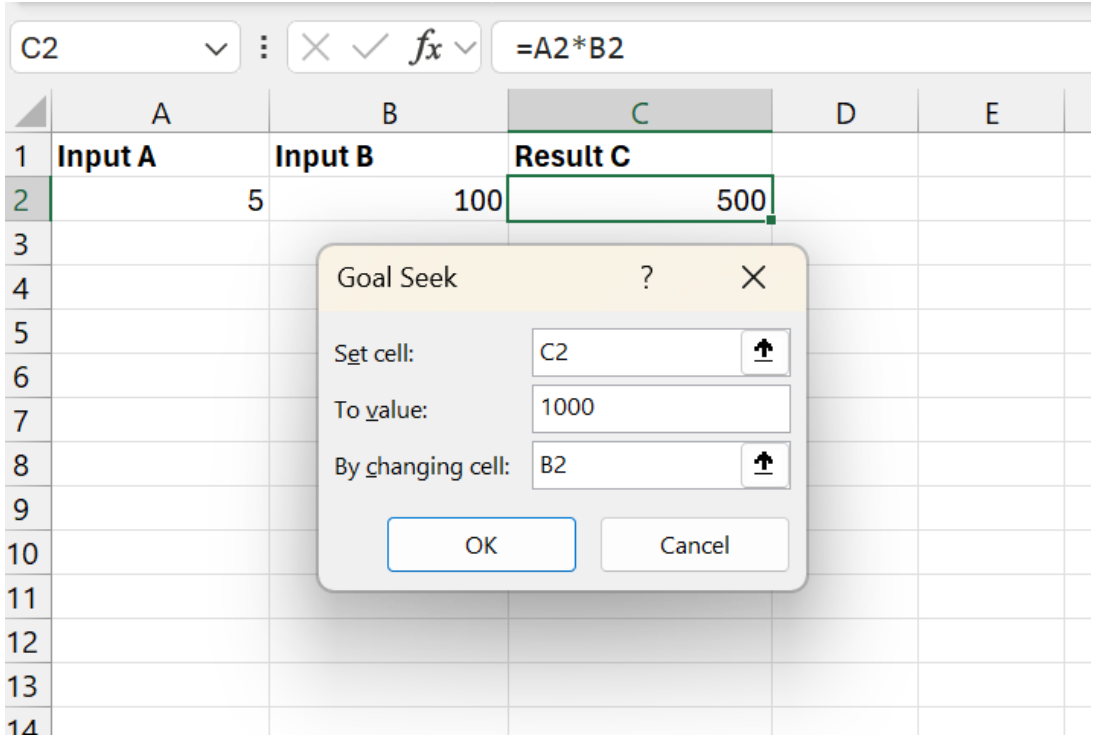
Adding Slicers

- Standard filters look boring.
- Slicers are visual, clickable buttons.
- PivotTable Analyze tab -> Insert Slicer.
- Click a button to watch your data update instantly.
- Turns a spreadsheet into a Dashboard.



Goal Seek

- Usually, you give Excel numbers to find the total.
- Goal Seek does the opposite.
- You give Excel the final answer you want.
- Excel changes an empty cell to make the math work.



The Target Marks Scenario

- You have 68 marks in Exam 1.
- You need a 75 average to pass the diploma.
- What mark do you need in Exam 2?
- Data -> What-If Analysis -> Goal Seek.
- Set cell to 75, by changing the blank Exam 2 cell!

Send Me Your Thoughts (Anonymously!)

Scan the QR code and share your questions or feedback

https://forms.gle/4WwKrDtcEDtG3X1BA

Thank You!